SAFEMAN · STUDENT MODULE

SM-PETRO

Petrochemical & Hydrogen Masterclass

SafeMan™ by Oryx Institute · safeman.oryxinstitute.org

Sector Masterclass · Pass mark 70% · 1 modules

Name _____

Student no. _____ Class _____



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WELCOME

SafeMan Petrochemical, Oil, Gas & Green Hydrogen Masterclass

SafeMan Petrochemical, Oil, Gas & Green Hydrogen Masterclass — Sector Masterclass. This SafeMan course is engineered for corporate audit readiness against: ATEX/IECEx zone practice; ISO 45001; Process Safety Management principles.

REGULATORY DISCLOSURE — READ FIRST

SafeMan™ is a proprietary inter-organizational safety management and operational competency benchmark developed by Oryx Institute (safeman.oryxinstitute.org). Courses are engineered in alignment with ISO 45001:2018, ISO 14001:2015, ISO 31000, ILO-OSH guidelines, and pre-referenced against Namibia Training Authority (NTA) Unit Standards and Namibian Government Notice 156 of 1997 for corporate audit readiness. SafeMan credentials verify industry competency and attendance between organizations and are non-NQA credit-bearing.

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How to use this module pack

Work through the modules in order. Each module states its focus and the key points you are expected to carry into practice. Diagrams referenced as web assets accompany the online reader at safeman.oryxinstitute.org. After the modules, sit the knowledge check: 6 questions, pass mark 70%, closed book.

MODULES

Module 1 — Explosive zones, PSM & hydrogen/ammonia

Zoned areas and ignition control, the pillars of Process Safety Management, and the specific behaviours of hydrogen and ammonia.

Key points

Zone 0/1/2 classification drives Ex-rated equipment; the permit and gas test gate every spark.

PSM pillars: process safety information, HAZOP, MOC, mechanical integrity, training, incident learning.

Hydrogen: wide flammability range, invisible flame, high leak rate — detectors and bond/earth discipline.

Ammonia: toxic and refrigerant-cold; water spray for knockdown, escape sets, and never entering a cloud.

Diagram: /assets/webp/sm-gas-detector.webp

ASSESSMENT

Knowledge check

The closed-book knowledge check draws 6 questions from the bank reproduced in Appendix A. Pass mark 70%. Your SafeMan credential is issued after the knowledge check, and remains valid for 24 months from issue.

APPENDIX A — QUESTION BANK

Knowledge check bank

Q1. Zone 1 is where an explosive atmosphere:

A. Never appears

B. Is likely in normal operation

C. Exists only in accidents

D. Is always present

Q2. HAZOP is a:

A. Pay system

B. Structured process hazard study using guide words

C. Fire drill

D. Audit clause

Q3. Hydrogen flames are:

A. Bright yellow

B. Nearly invisible in daylight

C. Always blue

D. Smoky

Q4. Management of Change (MOC) reviews:

A. Only budgets

B. Every technical/organizational change for new hazards

C. Only staff changes

D. Nothing

Q5. Ammonia release response:

A. Enter with dust mask

B. Avoid entry, use water spray knockdown and escape sets

C. Light a flare

D. Ventilate by fan

Q6. Ex-rated equipment exists to:

A. Save power

B. Prevent the equipment igniting the zone atmosphere

C. Reduce noise

D. Track inventory

Answer key

Q1=B Q2=B Q3=B Q4=B Q5=B Q6=B